

PAPER_385 — Canonical 7-System UQFF Parameter Registry

This paper establishes that registry — the authoritative configuration table for UQFF validation

Session: 104 (Complete Re-Analysis — full 18-field per-system registry not formalized in prior papers)

PAPER_385 — Canonical 7-System UQFF Parameter Registry

Source: grokshare11254865.txt, lines ~6700–6850 (confirmed 9400–10322 in main()) **Section:** MUGESystem struct initializations for all 7 canonical validation systems **Session:** 104 (Complete Re-Analysis — full 18-field per-system registry not formalized in prior papers) **CP4** **Class:** Canonical7SystemUQFFParameterRegistryCalculator (CP4 #36)

1. Overview

PAPER379 compared the compressed vs resonance MUGE outputs for all 7 canonical systems. PAPER377 noted the 18-field CSV format. But no paper explicitly documented all **18 parameters for each of the 7 systems** as a canonical reference registry.

This paper establishes that registry — the authoritative configuration table for UQFF validation that all C++ and Python implementations must match.

2. The 18-Field MUGESystem Parameter Set

The MUGESystem struct defines 18 fields for each astrophysical system:

Field	Symbol	Description	Units
1. name	—	System identifier	string
2. I	I	Electric current	A
3. A	A	Cross-sectional area	m ²
4. omega1	ω■	Upper frequency	rad/s
5. omega2	ω■	Lower frequency	rad/s
6. Vsys	V_sys	System volume	m ³
7. vexp	v_exp	Expansion velocity	m/s
8. t	t	System age	s
9. z	z	Redshift	—
10. ffluid	f_fluid	Fluid oscillation frequency	Hz
11. M	M	Total mass	kg
12. r	r	Characteristic radius	m
13. B	B	Magnetic field	T
14. Bcrit	B_crit	Critical magnetic field	T
15. rho_fluid	ρ_f	Fluid density	kg/m ³
16. g_local	g_local	Local gravitational surface acc.	m/s ²

Field	Symbol	Description	Units
17. M_DM	M_DM	Dark matter mass	kg
18. deltarhorho	$\delta\rho/\rho$	Fractional density contrast	—

3. Canonical Parameter Registry — All 7 Systems

System 1: SGR1745 — Magnetar SGR 1745-2900

Field	Value
I	1×10^{21} A
A	$3.142\times10^{\blacksquare}$ m ²
$\omega_{\blacksquare}$	1×10^{12} rad/s
$\omega_{\blacksquare}$	9.99×10^{11} rad/s
V_sys	4.189×10^{12} m ³
v_exp	1×10^3 m/s
t	3.799×10^{10} s
z	0.0009
f_fluid	$1.269\times10^{-1\blacksquare}$ Hz
M	2.984×10^{30} kg
r	$1\times10^{\blacksquare}$ m
B	1×10^{10} T
B_crit	1×10^{11} T
ρ_f	$1\times10^{1\blacksquare}$ kg/m ³
g_local	1.991×10^{12} m/s ²
M_DM	$1\times10^2\blacksquare$ kg
$\delta\rho/\rho$	0.1

System 2: Sag A — Sagittarius A (SMBH)

Field	Value
I	1×10^{23} A
A	2.813×10^{30} m ²
$\omega_{\blacksquare}$	1×10^{12} rad/s
$\omega_{\blacksquare}$	9.99×10^{11} rad/s
V_sys	$3.552\times10^{\blacksquare\blacksquare\blacksquare}$ m ³
v_exp	$5\times10^{\blacksquare}$ m/s
t	$3.786\times10^{1\blacksquare}$ s
z	0.0009
f_fluid	$3.465\times10^{-\blacksquare}$ Hz
M	$8.155\times10^3\blacksquare$ kg
r	1×10^{12} m
B	$1\times10^{-\blacksquare}$ T

Field	Value
B_crit	$1 \times 10^{-11} \text{ T}$
ρ_f	$1 \times 10^{-11} \text{ kg/m}^3$
g_{local}	$5.443 \times 10^2 \text{ m/s}^2$
M_DM	$1 \times 10^3 \text{ kg}$
$\delta\rho/\rho$	0.01

System 3: Tapestry — Tapestry of Blazing Starbirth

Field	Value
I	$1 \times 10^{22} \text{ A}$
A	$1 \times 10^3 \text{ m}^2$
ω_{I}	$1 \times 10^{12} \text{ rad/s}$
ω_{A}	$9.99 \times 10^{11} \text{ rad/s}$
V_sys	$1 \times 10^3 \text{ m}^3$
v_exp	$1 \times 10^3 \text{ m/s}$
t	$3.156 \times 10^{13} \text{ s}$
z	0.0
f_fluid	$1 \times 10^{-12} \text{ Hz}$
M	$1.989 \times 10^3 \text{ kg}$
r	$3.086 \times 10^1 \text{ m}$
B	$1 \times 10^{-11} \text{ T}$
B_crit	$1 \times 10^{-11} \text{ T}$
ρ_f	$1 \times 10^{-21} \text{ kg/m}^3$
g_{local}	$1.39 \times 10^{-1} \text{ m/s}^2$
M_DM	$1 \times 10^3 \text{ kg}$
$\delta\rho/\rho$	0.01

System 4: Westerlund 2 — Westerlund 2 Star Cluster

Field	Value
I	$1 \times 10^{22} \text{ A}$
A	$1 \times 10^3 \text{ m}^2$
ω_{I}	$1 \times 10^{12} \text{ rad/s}$
ω_{A}	$9.99 \times 10^{11} \text{ rad/s}$
V_sys	$1 \times 10^3 \text{ m}^3$
v_exp	$1 \times 10^3 \text{ m/s}$
t	$3.156 \times 10^{13} \text{ s}$
z	0.0
f_fluid	$1 \times 10^{-12} \text{ Hz}$
M	$1.989 \times 10^3 \text{ kg}$

Field	Value
r	$3.086 \times 10^{11} \text{ m}$
B	$1 \times 10^{-11} \text{ T}$
B_crit	$1 \times 10^{-11} \text{ T}$
ρ_f	$1 \times 10^{-21} \text{ kg/m}^3$
g_local	$1.39 \times 10^{-11} \text{ m/s}^2$
M_DM	$1 \times 10^3 \text{ kg}$
$\delta\rho/\rho$	0.01

Note: Tapestry and Westerlund 2 share identical parameters — they represent two systems in the same star-forming complex with equivalent physical scale.

System 5: Pillars of Creation

Field	Value
I	$1 \times 10^{21} \text{ A}$
A	$2.813 \times 10^{32} \text{ m}^2$
ω_1	$1 \times 10^{12} \text{ rad/s}$
ω_2	$9.99 \times 10^{11} \text{ rad/s}$
V_sys	$3.552 \times 10^{11} \text{ m}^3$
v_exp	$2 \times 10^3 \text{ m/s}$
t	$3.156 \times 10^{13} \text{ s}$
z	0.0
f_fluid	$8.457 \times 10^{-11} \text{ Hz}$
M	$1.989 \times 10^{32} \text{ kg}$
r	$9.46 \times 10^{11} \text{ m}$
B	$1 \times 10^{-10} \text{ T}$
B_crit	$1 \times 10^{-11} \text{ T}$
ρ_f	$1 \times 10^{-23} \text{ kg/m}^3$
g_local	$2.979 \times 10^{-10} \text{ m/s}^2$
M_DM	$1 \times 10^{32} \text{ kg}$
$\delta\rho/\rho$	0.05

System 6: Rings of Relativity — Rings of Relativity Gravitational Lens

Field	Value
I	$1 \times 10^{22} \text{ A}$
A	$1 \times 10^3 \text{ m}^2$
ω_1	$1 \times 10^{12} \text{ rad/s}$
ω_2	$9.99 \times 10^{11} \text{ rad/s}$
V_sys	$1 \times 10^{11} \text{ m}^3$
v_exp	$1 \times 10^3 \text{ m/s}$

Field	Value
t	$3.156 \times 10^1 \text{ s}$
z	0.01
f_fluid	$1 \times 10^1 \text{ Hz}$
M	$1.989 \times 10^3 \text{ kg}$
r	$3.086 \times 10^1 \text{ m}$
B	$1 \times 10^{-10} \text{ T}$
B_crit	$1 \times 10^1 \text{ T}$
ρ_f	$1 \times 10^{-2} \text{ kg/m}^3$
g_local	$1.391 \times 10^{-1} \text{ m/s}^2$
M_DM	$1 \times 10^3 \text{ kg}$
$\delta\rho/\rho$	0.02

System 7: Student's Guide — Student's Guide to the Universe (Cosmological)

Field	Value
I	$1 \times 10^2 \text{ A}$
A	$1 \times 10^2 \text{ m}^2$
ω	$1 \times 10^{12} \text{ rad/s}$
ω	$9.99 \times 10^{11} \text{ rad/s}$
V_sys	$1 \times 10^0 \text{ m}^3$
v_exp	$3 \times 10^1 \text{ m/s}$
t	$4.35 \times 10^1 \text{ s}$
z	0.0
f_fluid	$1 \times 10^{-1} \text{ Hz}$
M	$1 \times 10^3 \text{ kg}$
r	$1 \times 10^2 \text{ m}$
B	$1 \times 10^{-1} \text{ T}$
B_crit	$1 \times 10^{-1} \text{ T}$
ρ_f	$1 \times 10^{-2} \text{ kg/m}^3$
g_local	$6.67 \times 10^{-33} \text{ m/s}^2$
M_DM	$1 \times 10^3 \text{ kg}$
$\delta\rho/\rho$	0.001

4. CSV Format (18-Field Standard)

The canonical CSV header for UQFF system configuration files:

```
name,I,A,omega1,omega2,Vsys,vexp,t,z,ffluid,M,r,B,Bcrit,rho_fluid,g_local,M_DM,delta_rho_rho
sgr1745,1e21,3.142e8,1e12,9.99e11,4.189e12,1e3,3.799e10,0.0009,1.269e-14,2.984e30,1e4,1e10,1e11,1e15,1.991e12,1e28,0.1
sagA,1e23,2.813e30,1e12,9.99e11,3.552e45,5e6,3.786e14,0.0009,3.465e-8,8.155e36,1e12,1e-5,1e-4,1e-19,5.443e2,1e38,0.01
```

```
tapestry, 1e22, 1e35, 1e12, 9.99e11, 1e53, 1e4, 3.156e13, 0.0, 1e-12, 1.989e35, 3.086e17, 1e-9, 1e-8, 1e-21, 1.39e-15, 1e36, 0.01
westerlund, 1e22, 1e35, 1e12, 9.99e11, 1e53, 1e4, 3.156e13, 0.0, 1e-12, 1.989e35, 3.086e17, 1e-9, 1e-8, 1e-21, 1.39e-15, 1e36, 0.01
pillars, 1e21, 2.813e32, 1e12, 9.99e11, 3.552e48, 2e3, 3.156e13, 0.0, 8.457e-14, 1.989e32, 9.46e15, 1e-10, 1e-9, 1e-23, 2.979e-10, 1e32, 0.05
rings, 1e22, 1e35, 1e12, 9.99e11, 1e54, 1e5, 3.156e14, 0.01, 1e-9, 1.989e36, 3.086e17, 1e-10, 1e-9, 1e-28, 1.391e-14, 1e38, 0.02
student_guide, 1e24, 1e52, 1e12, 9.99e11, 1e80, 3e8, 4.35e17, 0.0, 1e-18, 1e53, 1e26, 1e-15, 1e-14, 1e-26, 6.67e-33, 1e53, 0.001
```

5. System Scale Spectrum

System	Class	r (m)	M (kg)	Physics Domain
SGR1745	Magnetar	1×10^4	2.984×10^{30}	NS surface gravity
Sag A*	SMBH	1×10^{12}	8.155×10^3	Galactic center
Pillars	GMC	9.46×10^1	1.989×10^{32}	Stellar nursery
Tapestry	LMC-scale	3.086×10^1	1.989×10^3	Star-forming complex
Westerlund 2	Cluster	3.086×10^1	1.989×10^3	OB star cluster
Rings	Lens	3.086×10^1	1.989×10^3	Gravitational lens
Student's Guide	Cosmological	1×10^2	1×10^3	Observable universe

The 7 systems span **22 orders of magnitude** in radius (10^4 m to 10^{12} m) and **23 orders in mass** (10^{30} kg to 10^3 kg), making this the most comprehensive UQFF multi-scale validation suite in the codebase.

6. Canonical Computed Results Summary

For reference — results from PAPER379 (*full comparison*) and PAPER382/384 (per-term):

System	Compressed MUGE (m/s^2)	Resonance MUGE (m/s^2)	Ratio
SGR1745	1.782×10^3	1.773×10^3	$10^{0.001}$
Sag A*	2.966×10^3	4.105×10^2	$\sim 10^{-0.5}$
Tapestry	$\sim \text{GM}/r^2$	fluid-dominated	converge
Westerlund 2	$\sim \text{GM}/r^2$	fluid-dominated	converge
Pillars	$\sim \text{GM}/r^2$	fluid-dominated	converge
Rings	$\sim \text{GM}/r^2$	fluid-dominated	converge
Student's Guide	$\sim \text{GM}/r^2$	fluid-dominated	converge

7. References Within Codebase

PAPER_371: Resonance MUGE framework — 12-term formula

PAPER_372: Compressed MUGE framework — 8-term formula

PAPER_379: Dual-model 7-system comparison table

PAPER_377: `load_muge_systems()` CSV parser (18-field format)

`WormholeMUGETermImplSafetyCalculator` (CP4 #26): 7-system dict

MUGESuperconductive12TermResonanceCalculator (CP4 #14): uses sagA_dataset

*Source: grokshare11254865.txt lines ~6700–6850 + lines ~9400–10322 (main() C++ impl.) | Session 104 |
First formal 18-field canonical parameter registry for all 7 UQFF validation systems*

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